

Syllabus:

(MT - 173)

Course Content

1. Functions
2. Differential Calculus
3. Integral Calculus
4. Vector Calculus
5. Complex numbers.

Recommended Books

1. Engineering Mathematics Anthony Goff
2. Calculus Theory & Finney
3. Calculus & Analytical Geometry Howard Arto

Sessional marks distribution

1. Mid term examination 8th week (20 marks)
9, 10, 11, 12 → 5 star high
2. Assignments 7th week, 13th week (20 marks)
Lecture compilation
(L.O + Attendance + Bonus)

Includes CFP (complex Engineering Problem) → 3 marks

Finding solution, result, conclusion interpretation. marks

CLO2 weightage in Ass 2 → 3 marks, CLO1 → CFP

CLO1 weightage in Ass 1 → 7 marks, CFP → 3 marks

∴ Assignment 1 & CLO2 marks

Surface $\rightarrow$ 2 dimension figure
Solid $\rightarrow$ 3 dimension figure

Date _____

Functions

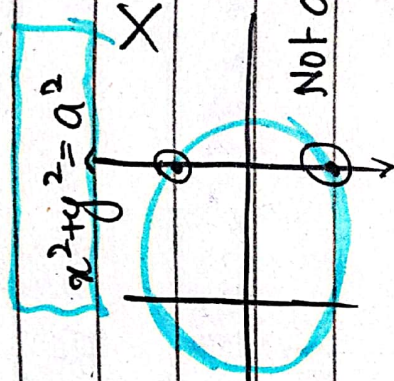
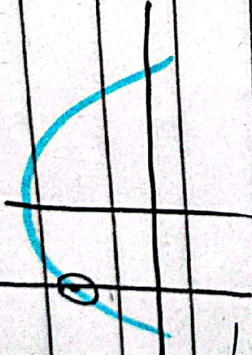
Vertical line agr 1 point p cut kre
to function otherwise not a function.

eg:

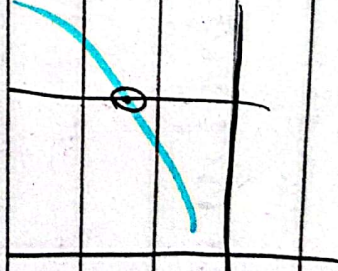
$$y = \sqrt{a^2 - x^2}$$

$\rightarrow$ is a function

(sq. root function)



Not a function



$e^x \rightarrow$ exponential function

means bhot rapid rise krta.
science m y growth &
decay curve kehlete.

$$x^{2/3} + y^{2/3} = a^{2/3}$$



Astroid $\rightarrow$ is not a function.

point $\rightarrow$ plot kate ; graph $\rightarrow$ trace kate.

Date _____

Piece Wise Defined Function.

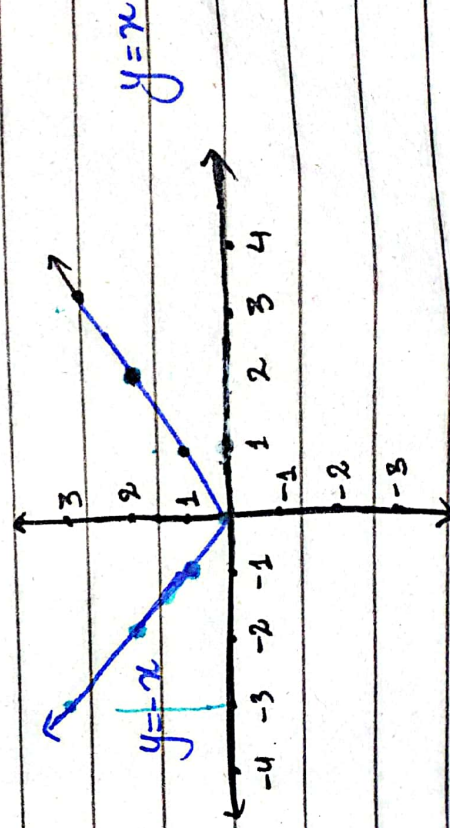
In mathematics a piecewise defined function is a function defined by multiple sub function and each subfunction applies to a diff. interval in the domain. We say that it is just a function made up of multiple function.

OR:

If piecewise defined function is a function whose domain is divided into parts & each part is defined by a different function rule.

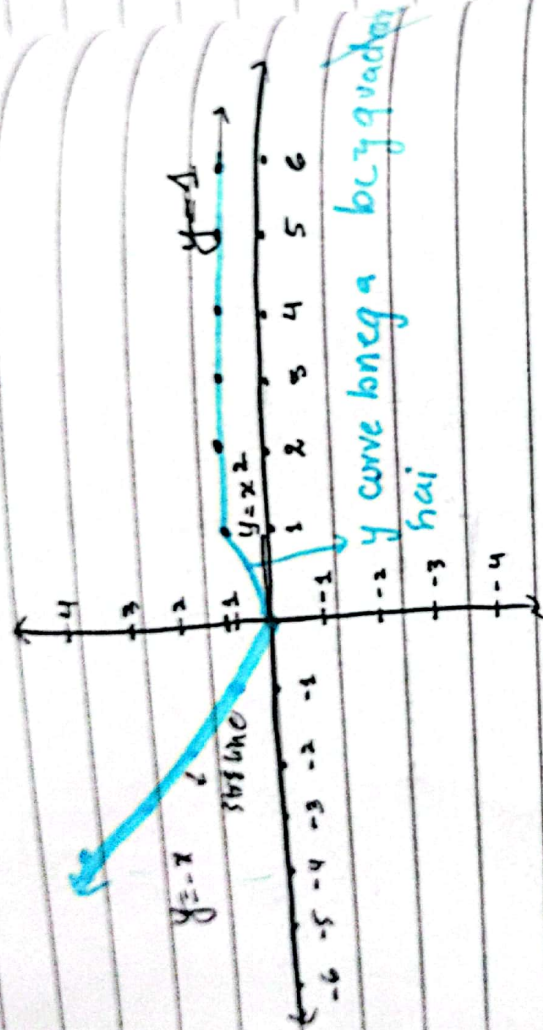
Ex: GRAPH THE FUNCTIONS:

$$(i) \quad y = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$



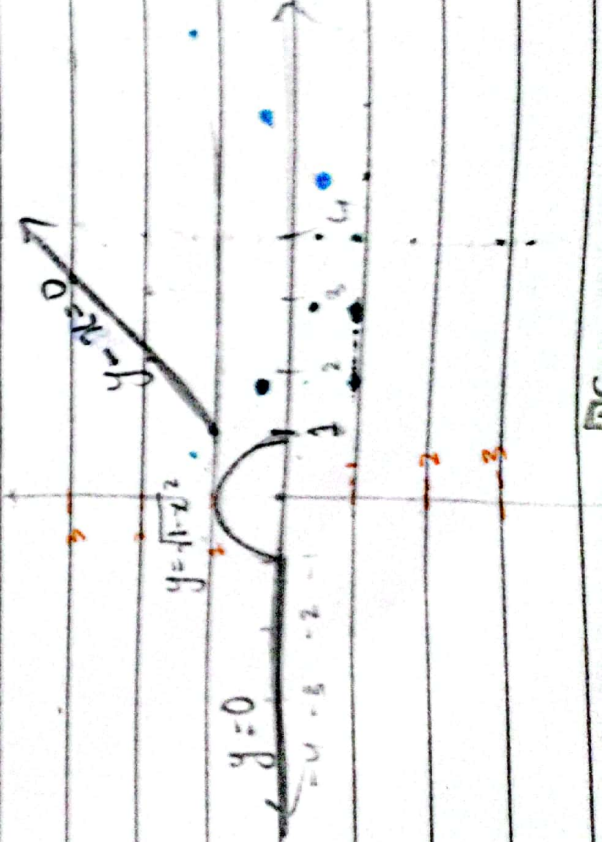
Date _____

$$(ii) f(x) = \begin{cases} -x & x < 0 \\ x^2 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$



$$f(t) = \begin{cases} 0 & t < 0 \\ t & 0 \leq t \leq 1 \\ 2t & t > 1 \end{cases}$$

$$(iii) f(x) = \begin{cases} 0 & x \leq -1 \\ \sqrt{1-x^2} & -1 < x < 1 \\ x & x \geq 1 \end{cases}$$



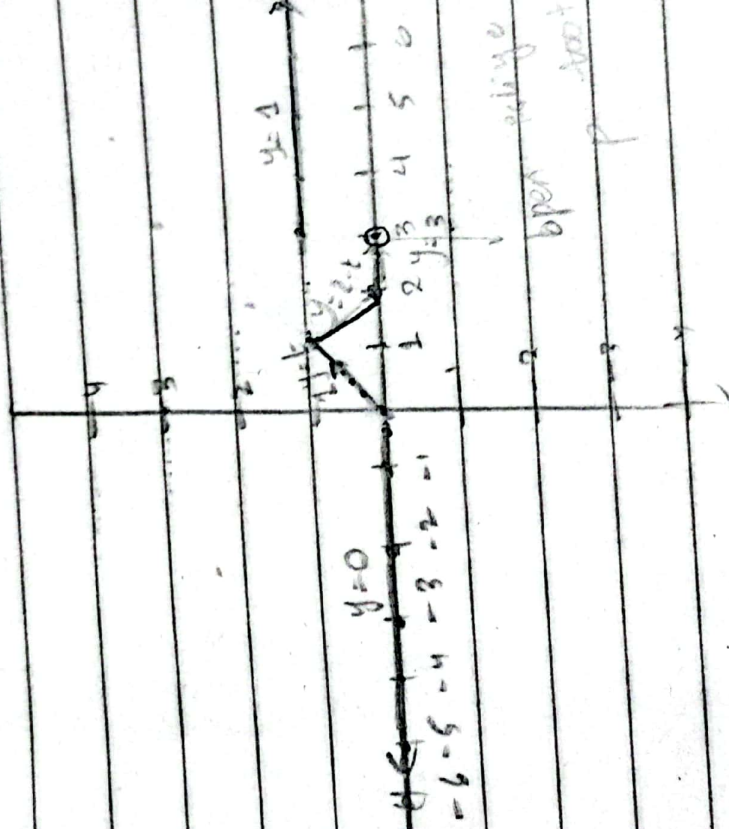
Greater

The graph
function
(less
real
[x])

0.1/16

Date:

$$f(t) = \begin{cases} 0 & t < 0 \\ 1 & 0 \leq t < 1 \\ 2 & 1 \leq t < 2 \\ 0 & 2 \leq t < 3 \\ 1 & t \geq 3 \end{cases}$$



Greatest Integer Function: (STEP or FLOOR FUNCTION)

The greatest integer function is a function that results in the integer (less than or equal to) to a given real number.
 $[x] \rightarrow$ The largest integer less than or equal to x .

Indet. a mivede formis je mag, no

1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100

age k₁₀₀ function kr

ajaye to home
limit kna

There are 7 mathematics

$$\frac{0.22}{0.22} \times 0.22 = 0.22$$

Evaluate the km

$$1) \lim_{x \rightarrow 5} \frac{5}{x-5}$$

$$1) \lim_{x \rightarrow 0} \frac{5}{x-5} \quad 2) \lim_{x \rightarrow 9} \frac{x-9}{x-9}$$

4) $\lim_{x \rightarrow -1} \left(\frac{x^3 + x^2}{x^2 - 1} \right)$

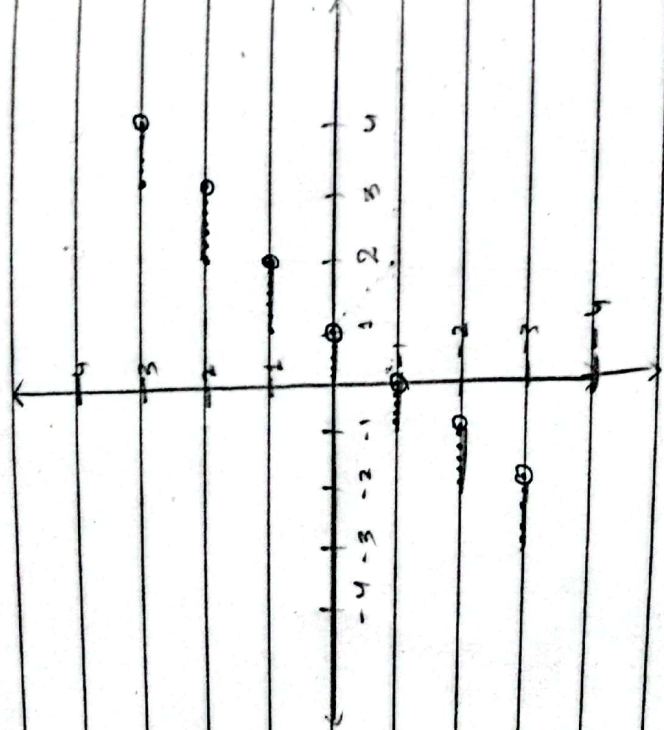
$$c) \lim_{x \rightarrow 100} [\sqrt{x^2 + 1} - x]$$

8) $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{\sin^2 x}$ d) l

$$1.27 \lim_{x \rightarrow 0} \frac{x}{\tan x}$$

$$13) \lim_{x \rightarrow 0} (1 + \dots)$$

$$(14) \lim_{p \rightarrow 0} (1 + \dots)$$



and the greatest integer function is continuous at all points except the integers.

Domain $\rightarrow$ All real values
Range $\rightarrow$ Set of integers

Date _____

1.5.15.3

ags kisi function ki indeterminate form
aaye to hum us function ko
limit kya hote hai

There are 7 indeterminate form in
mathematics

$$\frac{0}{0}, \frac{\infty}{\infty}, \frac{0 \cdot \infty}{\infty}, \frac{\infty - \infty}{\infty}, \frac{0^\circ, \infty^\circ, 0^\circ, 1^\circ}{\infty}$$

Evaluate the limits of the following

$$1) \lim_{x \rightarrow 0} \frac{5}{x-5} \quad 2) \lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3}}{x} \quad 3) \lim_{x \rightarrow 7} \frac{x^2 - 49}{x-7}$$

$$4) \lim_{x \rightarrow -1} \left(\frac{x^3 + x^2}{x^2 - 1} \right) \quad 5) \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x-2}$$

$$6) \lim_{x \rightarrow \infty} [\sqrt{x^2 + x + 1} - x] \quad 7) \lim_{x \rightarrow \infty} x \sin x \quad \frac{1}{x}$$

$$8) \lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{\sin^2 x} \quad 9) \lim_{x \rightarrow e} \frac{\ln x - 1}{x - e} \quad 10) \lim_{\theta \rightarrow 0} \frac{1 - \cos 2\theta}{\theta^2}$$

$$11) \lim_{x \rightarrow 0} \frac{x}{\tan x} \quad 12) \lim_{x \rightarrow 0} \frac{e^x - 3^x - 2^x + 1}{x^2}$$

$$13) \lim_{x \rightarrow 0} (1 + 3x)^{2/x}$$

$$14) \lim_{p \rightarrow 0} \left(1 + \frac{p}{5} \right)^{5/p} \quad 15) \lim_{x \rightarrow 0} \frac{e^{-2x} - e^{-x}}{x}$$

Date _____

Theorems

$$1) \lim_{x \rightarrow +\infty} (e^x) = \infty$$

$$2) \lim_{x \rightarrow -\infty} (e^x) = \lim_{x \rightarrow -\infty} \left(\frac{1}{e^{-x}} \right) = 0$$

$$3) \lim_{x \rightarrow \pm \infty} \left(\frac{a}{x} \right) = 0$$

$$4) \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

$$5) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$$

$$6) \lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$7) \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log a$$

$$8) \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = 1$$

Limit

limit of a function
no. of the
to "L" a
close en
Mathematical

kissi function
krte jo y
continuous h

Continuity

A function f is said to be continuous at a point x=a if the following conditions are satisfied:

(i) f(a) is defined
(ii) $\lim_{x \rightarrow a} f(x)$

(iii) $\lim_{x \rightarrow a} f(x) = f(a)$

If one or more of these conditions are not satisfied, then the function is said to be discontinuous at point x=a.

Limit:

limit of a function at point a is the no. "L" such that the values of the function get close to "L" as long as x becomes close enough to the point "a".
Mathematically, limit $\lim_{x \rightarrow a} f(x) = L$

Kissi function Ki limit ussing find krte j b y check krna function continuous hai ya discontinuous.

Continuity Criterion

A function is said to be continuous at a number a . If the following conditions are satisfied

(i) $f(a)$ is defined

(ii) $\lim_{x \rightarrow a} f(x)$ exists

(iii) $\lim_{x \rightarrow a} f(x) = f(a)$

If one or more of these three conditions failed to hold at a then the function is said to be discontinuous at point "a".

Date _____

Q1) Determine whether the function is continuous at $x=2$.

$$f(x) = \begin{cases} x^2 - 4 & \text{when } x \neq 2 \\ 3 & \text{when } x = 2 \end{cases}$$

(1) i)

$$f(2) = 3$$

finding limit

$$\lim_{x \rightarrow 2} (x^2 - 4) = \lim_{x \rightarrow 2} (x-2)(x+2) = 0 \cdot 4 = 0$$

$$f(x) = \begin{cases} x^2 - 4 & x \neq 2 \\ 4 & x = 2 \end{cases}$$

Since $f(2) = 0 \neq 4$

Therefore, the function is not continuous at $x=2$.

$$f(x) = \begin{cases} 2+x & \text{when } x < 3 \\ 5-2x & \text{when } x > 3 \end{cases}$$

Q2) function $f(x)$

Finding Right Hand Limit $\lim_{x \rightarrow 3^-}$

Discuss the continuity & discontinuity of the function at $x=2$

$$f(x) = \begin{cases} -x^4 + 3 & x \leq 2 \\ x^2 + 9 & x > 2 \end{cases}$$

$$= \lim_{x \rightarrow 3^+}$$

Since $\lim_{x \rightarrow 3} f(x)$

does not exist at $x=3$

(11.1)

$$f(2) = 3.$$

Finding limit at $x \rightarrow 2$.

$$f(x) = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)}$$

$$= (2+2)$$

$$= 4$$

Since $f(2) \neq \lim_{x \rightarrow 2}$ therefore the

function is discontinuous at $x=2$.

$$Q2) \quad f(3) = 5 - 2(3) = 5 - 6 = -1.$$

Finding Right hand and left hand limits

$$= \lim_{x \rightarrow 3^-} (2+x) = 2+3 = 5$$

$$= \lim_{x \rightarrow 3^+} 5 - 2(x) = 5 - 2(3) = -1$$

Since $\lim_{x \rightarrow 3^-} \neq \lim_{x \rightarrow 3^+}$ therefore, limit

does not exist. and hence, is discontinuous at $x=3$.

Date _____

Discuss the continuity of the function

at $x=3$

$$(i) f(x) = \begin{cases} x^2 - 9 & x \neq 3 \\ x - 3 & x = 3 \end{cases}$$

$$(ii) f(x) = \begin{cases} x - 1 & x < 3 \\ 2x + 1 & x \geq 3 \end{cases}$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

In general;

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

(successive)
 n^{th}

Evaluate the forms

$$1) \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

L-Hopital theorem is applicable to these two forms $\infty \cdot 0$ or $0 \cdot \infty$

Indeterminate forms: $0, \infty, 0 \cdot \infty, \infty \cdot 0$

RECOMMENDATIONS

If the above indeterminate forms occur in the calculation of limit of the function, then we apply L-Hopital Theorem

Method for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ forms: Then LHT says that

$$\lim_{x \rightarrow a} f(x) = 0 \text{ or } \lim_{x \rightarrow a} g(x) = 0$$

$$\lim_{x \rightarrow a} f(x) = \infty \text{ or } \lim_{x \rightarrow a} g(x) = \infty$$

$$= \lim_{x \rightarrow 0} \frac{1 + \cos x}{x}$$

$$3) \lim_{x \rightarrow 0} \frac{e^x}{x}$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

In general; LHT says that

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f^{(n)}(x)}{g^{(n)}(x)}$$

(successive differentiation $\rightarrow$ jo wo n^{th} order ki hif jarka ho).

Evaluate the following indeterminate forms

$$1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$$

$$2) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} = \lim_{x \rightarrow 0} \frac{0 + \sin x}{6x}$$

$$= \lim_{x \rightarrow 0} \frac{0 + \cos x}{6} = \frac{1}{6}$$

$$3) \lim_{x \rightarrow 0} \frac{e^x - (x+1)}{x(e^x - 1)} = \lim_{x \rightarrow 0} \frac{e^x - 1 + 0}{x(e^x - 1)}$$

Date _____

$$\lim_{x \rightarrow 0} \ln \cos x \quad (\text{Ans: } 1/9)$$

$$\lim_{x \rightarrow 0} \frac{e^x - 2 \cos x + e^x}{x \sin x} \quad (\text{Ans: } 2)$$

$$\lim_{x \rightarrow 0} \ln(1-x^2) \quad (\text{Ans: } 2)$$

$$\lim_{x \rightarrow 0} \frac{\ln x^2}{\cot x^2} \quad (\text{Ans: } 0)$$

The limit of the given function is indeterminate form of the type $\frac{0}{0}$.

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$$

Applying LHT;

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} \quad \left(\frac{0}{0} \right)$$

Applying LHT;

$$\lim_{x \rightarrow 0} \frac{0 + \sin x}{6x} \quad \left(\frac{0}{0} \right)$$

Applying LHT;

$$\lim_{x \rightarrow 0} \frac{\cos x}{6}$$

$$= \frac{1}{6} \text{ ans}$$

-8 (03)

The limit of indeterminate form

Solution:

$$\lim_{x \rightarrow 0} \frac{e^x - (x+1)}{x(e^x - 1)}$$

Applying LHT

$$\lim_{x \rightarrow 0} \frac{e^x}{xe^x + e^x}$$

Applying LHT

$$= \frac{1}{0+1+1}$$

$$4) \lim_{x \rightarrow 0} \frac{\ln \cos x}{\ln \cos 3x}$$

The limit of the type

$$\lim_{x \rightarrow 0} \frac{\ln \cos x}{\ln \cos 3x}$$

Applying LHT

$$= \frac{1}{\cos x}$$

8 (03) 3

The limit of the function is indeterminate form of the type $(\frac{0}{0})$.

Solution:

$$\lim_{x \rightarrow 0} \frac{e^x - (x+1)}{x(e^x - 1)}$$

Applying LHT;

$$\lim_{x \rightarrow 0} \frac{e^x - (1+0)}{xe^x + (e^x - 1)}$$

Applying LHT;

$$\lim_{x \rightarrow 0} \frac{e^x}{xe^x + e^x + e^x}$$

Applying limit

$$= \frac{1}{0+1+1} = \frac{1}{2} \text{ Ans!}$$

$$4) \lim_{x \rightarrow 0} \frac{\ln \cos x}{\ln \cos 3x}$$

The limit of the function is indeterminate form of the type $(\frac{0}{0})$

$$\lim_{x \rightarrow 0} \frac{\ln \cos x}{\ln \cos 3x} = \lim_{x \rightarrow 0} \frac{\tan x}{3 \tan 3x}$$

Applying LHT;

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x}{9 \sec^2 3x}$$

Applying LHT

$$= \frac{\sin x}{\cos x} \div \frac{3 \sin 3x}{\cos 3x}$$

$$= \frac{1}{3} \cdot \frac{\cos 3x}{\cos x} = \frac{1}{3}$$

Date _____

type (0/0) -

$$\lim_{x \rightarrow 0} \frac{e^x - 2\cos x + e^{-x}}{x \sin x}$$

applying LHT:

$$= \lim_{x \rightarrow 0} \frac{e^x + 2\sin x - e^{-x}}{x \cos x + \sin x} \quad (0/0)$$

applying LHT

$$= \lim_{x \rightarrow 0} \frac{e^x + 2\cos x + e^{-x}}{-x \sin x + \cos x + \cos x}$$

$$= \frac{1 + 2 + 1}{0 + 1 + 1} = \frac{4}{2} = 2 \text{ Ans}$$

Q. 7

type (0/0)

$$\lim_{x \rightarrow 0} \frac{\ln(1 - \cos^2 x)}{\ln \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - 2\cos x \sin x}{(1 - \cos^2 x)}$$

$$= \lim_{x \rightarrow 0} \frac{-2 \sin 2x}{\sin^2 x} \quad (0/0)$$

applying L.H.T

$$= \lim_{x \rightarrow 0} \frac{-4 \cos 2x}{\cos^2 x} =$$

BC

Q no: 2) Method
or (∞)

To evaluate $\lim_{x \rightarrow a}$

$$\lim_{x \rightarrow a} f(x) = 0 \&$$

$$\text{or } \lim_{x \rightarrow a} f(x) = \infty$$

then write;

$$\lim_{x \rightarrow a} f(x)g(x)$$

$$\text{or } \lim_{x \rightarrow a} (g(x) \cdot f(x))$$

Q no: 2) Method for $(0 \cdot \infty)$ or $(\infty \cdot 0)$ form

To evaluate $\lim_{x \rightarrow a} f(x) \cdot g(x)$; when

$$\lim_{x \rightarrow a} f(x) = 0 \text{ \& } \lim_{x \rightarrow a} g(x) = \infty.$$

or

$$\lim_{x \rightarrow a} f(x) = \infty \text{ \& } \lim_{x \rightarrow a} g(x) = 0$$

then write;

$$\lim_{x \rightarrow a} f(x)g(x) = \lim_{x \rightarrow a} \frac{f(x)}{\frac{1}{g(x)}} \quad \text{or} \quad \lim_{x \rightarrow a} \frac{g(x)}{\frac{1}{f(x)}}$$

or

$$\lim_{x \rightarrow a} (g(x) \cdot f(x)) = \lim_{x \rightarrow a} \frac{g(x)}{\frac{1}{f(x)}} \quad \text{or} \quad \lim_{x \rightarrow a} \frac{f(x)}{\frac{1}{g(x)}}$$

Ex. 2) Evaluate the following indeterminate forms.

1) $\lim_{x \rightarrow 0} x \ln(\tan x)$ {Ans: 0}

2) $\lim_{x \rightarrow 0} \sin x \ln x$ {Ans: 0}

3) $\lim_{x \rightarrow 0} x \tan\left(\frac{\pi}{2} - x\right)$ {Ans: 2}

4) $\lim_{x \rightarrow \pi/2} (1 - \sin x) \tan x$ {Ans: 0}

The limit of the function is indeterminate form of the type $(0 \cdot \infty)$.

$\lim_{x \rightarrow 0} x \ln(\tan x)$ $(0 \cdot \infty)$

$= \lim_{x \rightarrow 0} \frac{\ln(\tan x)}{1/x}$ $\left(\frac{\infty}{\infty}\right)$

By applying L.H.T,

$= \lim_{x \rightarrow 0} \frac{\tan x - \sec^2 x}{1/x^2} = \lim_{x \rightarrow 0} \frac{-\sec^2 x \cdot x^2}{\tan x}$

Ex. $\lim_{x \rightarrow 0} \frac{-x^2}{\tan x} = \lim_{x \rightarrow 0} \frac{-x^2 \sec^2 x}{\tan x}$ or $\lim_{x \rightarrow 0} \frac{-x^2}{\tan x}$

By applying L.H.T

$= \lim_{x \rightarrow 0} \frac{-2x}{\tan^2 x}$

$= \lim_{x \rightarrow 0} \frac{-2x}{x^2}$

$= \lim_{x \rightarrow 0} \frac{-2}{x}$

$= 0$

Ex. The limit of $\frac{\sin x}{x}$ as $x \rightarrow 0$

$\lim_{x \rightarrow 0} \frac{\sin x}{x}$

$= \lim_{x \rightarrow 0} \frac{\cos x}{1}$

$= \lim_{x \rightarrow 0} \cos x$

By applying L.H.T

$= \lim_{x \rightarrow 0} \frac{-\sin x}{-1}$

$= \lim_{x \rightarrow 0} \sin x$

$= 0$

$= \lim_{x \rightarrow 0} x$

By applying LHT

$$= \lim_{x \rightarrow 0} (-2 \sec^2 x \cdot x + \dots)$$

$$= \lim_{x \rightarrow 0} \frac{-2x}{\cos^2 x \cdot \sec^2 x}$$

$$= -2(0)$$

$$= 1 - 0$$

$$= 0 \text{ Ans!}$$

Q22 The limit $\left(\frac{\pi}{2} - x\right) \dots$ type $(0 \cdot \infty)$

$$\lim_{x \rightarrow 0} \sec x \cdot \ln x = \lim_{x \rightarrow 0} \frac{1}{\cos x} \cdot \ln x \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{\cos x} \cdot \ln x \quad \text{By applying LHT} = \lim_{x \rightarrow 0} \frac{-\frac{1}{x}}{\cos x}$$

$$= \lim_{x \rightarrow 0} \frac{-\cos x}{x^2} \quad \left(\frac{0}{0}\right)$$

By applying LHT

$$= \lim_{x \rightarrow 0} \frac{\sin x}{2x} \quad \text{LHT}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x}{2} \quad \text{Applying LHT}$$

By 2nd (applying limit), we get:

$$= \lim_{x \rightarrow 0} \frac{\cos x}{2} \quad \left(\frac{0}{0}\right)$$

$$= 0$$

$$= \lim_{x \rightarrow 0} \frac{2}{2} = 1$$

$$2 \cos x$$

(1) $\lim_{x \rightarrow 0} \frac{1}{\sec^2 x} = 0$ sec² form

Q) The limit of the function is indeterminate form of the type $(0 \cdot \infty)$

$$\lim_{x \rightarrow \pi/2} (1 - \sin x) \tan x \quad (0 \cdot \infty)$$

$$= \lim_{x \rightarrow \pi/2} \frac{(1 - \sin x) = \frac{1 - \sin x}{\cot x}}{\tan x}$$

By applying LHT

$$= \lim_{x \rightarrow \pi/2} \frac{-\cos x}{-\operatorname{cosec}^2 x}$$

$$= \lim_{x \rightarrow \pi/2} \cos x \times \sin^2 x$$

By applying limit

$$= \cos(\pi/2) \cdot \sin^2(\pi/2)$$

$$= 0 \times 1 = 0 \quad \text{Ans!}$$

Q3) The limit

$$\lim_{x \rightarrow 0} x \tan\left(\frac{1}{x}\right)$$

$$= \lim_{x \rightarrow 0}$$

$$= \lim_{x \rightarrow 0}$$

$$= \lim_{x \rightarrow 0}$$

applying

$$= \lim_{x \rightarrow 0}$$

applying

$$= \frac{1}{1}$$

$$= \frac{1}{1}$$

$$= \frac{1}{1}$$

Date

(0.00)

Q3) The limit $\dots\dots\dots$ (0.00)

$\lim_{x \rightarrow 0}$

$$= \lim_{x \rightarrow 0} \frac{\cot x}{1/x} \quad \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\cos x \csc^2 x}{1/x^2} \quad (\text{By applying LHT})$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x} \quad (\%)$$

Applying LHT

$$= \lim_{x \rightarrow 0} \frac{2x}{2 \cos x \sin x} \quad (\%)$$

Applying LHT

$$= \lim_{x \rightarrow 0} \frac{1}{\cos^2 x - \sin^2 x}$$

$$= \frac{1}{1-0} \quad (\text{By applying limit})$$

$$= 1$$

$$= 1$$

Date _____

ex. 1.1.1

$$\lim_{x \rightarrow 0} x \tan\left(\frac{\pi}{2} - x\right)$$

$$= \lim_{x \rightarrow 0} \frac{x}{\tan\left(\frac{\pi}{2} - x\right)} \quad (\frac{0}{0})$$

$$= \lim_{x \rightarrow 0} \frac{1}{\cot\left(\frac{\pi}{2} - x\right)}$$

Applying LHT,

$$\lim_{x \rightarrow 0} \frac{1}{-\cot^2\left(\frac{\pi}{2} - x\right)(-1)}$$

$$\lim_{x \rightarrow 0} \frac{1}{\cot^2\left(\frac{\pi}{2} - x\right)}$$

$$\lim_{x \rightarrow 0} \sin^2\left(\frac{\pi}{2} - x\right)$$

$$= 1.$$

Informal

The exponential function is written as:

- 1) $e^x = \exp(x)$
- 2) $e^{x+y} = \exp(x+y)$
- 3) $e^{\tan x} = \exp(\tan x)$

Method for

To evaluate,

$$(i) \lim_{x \rightarrow a} f(x) = 0$$

$$(ii) \lim_{x \rightarrow a} f(x) = \infty$$

$$(iii) \lim_{x \rightarrow a} f(x) = 1$$

Then the given,

$$\lim_{x \rightarrow a} [f(x)]^{g(x)}$$

or

$$= \exp \left[\lim_{x \rightarrow a} g(x) \log f(x) \right]$$

The RHS of

$\infty \cdot 0$ form which method.

Information:

The exponential function can also be

written as:

$$1) e^x = \exp(x)$$

$$2) e^{x+y} = \exp(x+y)$$

$$3) e^{\tan x} = \exp(\tan x)$$

Method for $0^0, \infty^0, 1^\infty$ forms:

when

To evaluate $\lim_{x \rightarrow a} [f(x)]^{g(x)}$ & $\lim_{x \rightarrow a} g(x) = 0$

$$(i) \lim_{x \rightarrow a} f(x) = 0$$

$$(ii) \lim_{x \rightarrow a} f(x) = \infty \text{ \& } \lim_{x \rightarrow a} g(x) = 0$$

$$(iii) \lim_{x \rightarrow a} f(x) = 1 \text{ \& } \lim_{x \rightarrow a} g(x) = \infty$$

Then the given limit can be written as

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = \lim_{x \rightarrow a} \exp[\ln f(x) g(x)]$$

$$= \lim_{x \rightarrow a} \exp[g(x) \ln f(x)]$$

or

$$= \exp \left[\lim_{x \rightarrow a} g(x) \ln f(x) \right]$$

The RHS of the above equation is $0 \cdot \infty$ or $\infty \cdot 0$ form what can be evaluated by previous method.

4) Logarithmic growth is very slow & logarithm while exponential is rapid. $\rightarrow$ Physics term.

Date _____

Let $\lim_{x \rightarrow a} g(x) \ln f(x) = L$

Then $\lim_{x \rightarrow a} f(x)^{g(x)} = \exp(L)$

or $\lim_{x \rightarrow a} f(x)^{g(x)} = e^L$

-d part
Solution

Ex: Evaluate the following indeterminate form

1) $\lim_{x \rightarrow 0} (1+x)^{1/x}$ Ans: e

2) $\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\tan x}$ Ans: 1

3) $\lim_{x \rightarrow \pi/2} (1-\sin x)^{\cos x}$ Ans: 1

4) $\lim_{x \rightarrow 0} (\cot x)^{\sin x}$ Ans: 1

5. $\lim_{x \rightarrow 0} (1+\sin x)^{\cot x}$ Ans: e

6. $\lim_{x \rightarrow 5} (x-5)^{x-5}$ Ans: 1

7. $\lim_{x \rightarrow 1} (1-x^2)^{\frac{1}{\ln(1-x)}}$ Ans: e

$$\lim_{x \rightarrow 0} \exp[\ln(1+x)]$$

$$\exp \left[\lim_{x \rightarrow 0} \left\{ \frac{\ln(1+x)}{x} \right\} \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \frac{1}{1+x} \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \left(\frac{1}{1} \right) \right]$$

$$= \exp(1)$$

$$= e^1 = e$$

$$= \exp \left[\frac{1}{1} \right]$$

part i

Solution

$$\lim_{x \rightarrow 0} \exp[\ln(1+x)^{1/x}] \quad (\infty \cdot 0)$$

$$\exp \left[\lim_{x \rightarrow 0} \left\{ \frac{1}{x} \ln(1+x) \right\} \right] \quad (\infty \cdot 0)$$

$$= \exp \left[\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \right] \quad (0/0)$$

$$= \exp \left[\lim_{x \rightarrow 0} \frac{\frac{1}{1+x} (1)}{1} \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \left(\frac{1}{1+x} \right) \right]$$

$$= \exp(1)$$

$$= e^1 = e$$

$$= \exp \left[\lim_{x \rightarrow 0} \frac{-\frac{2x}{(1+x)^2} (1+x)}{-1} \right]$$
$$= \exp \left[\lim_{x \rightarrow 0} \frac{-2x}{(1+x)} \right] = e$$

Date _____

part ii b

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\tan x} \quad (\infty^0)$$

$$= \exp \left[\lim_{x \rightarrow 0} \left[\tan x \ln \left(\frac{1}{x} \right) \right] \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \left[\ln \left(\frac{1}{x} \right) \cdot \frac{1}{\tan x} \right] \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \left[\frac{\ln \frac{1}{x}}{\frac{1}{\tan x}} \right] \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \left[\frac{\ln \frac{1}{x}}{\frac{1}{\tan x}} \right] \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \left[\frac{\ln \frac{1}{x}}{\frac{1}{\tan x}} \right] \right]$$

$$= \exp \left[\lim_{x \rightarrow 0} \left[\frac{\ln \cos x \cdot \sin x}{1} \right] \right]$$

$$= \exp [0]$$

$$= 1$$

Method for

In this indeterminate function or by the can be converted in

Ex. Evaluate the following forms:

$$1) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$$

$$2) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$$

$$3) \lim_{x \rightarrow 0} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right)$$

$$4) \lim_{x \rightarrow 4} \left(\frac{1}{\ln(x-3)} - \frac{1}{x-4} \right)$$

part ii

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$$

The limit of the form of the

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\tan x} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$$

Method for $\infty-\infty$ form

In this indeterminate form: Simplify the function or by taking L.C.M, it can be converted into $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.
 Ex. Evaluate the following indeterminate forms.

$$1) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right) \quad \text{Ans: } 0$$

$$2) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right) \quad \text{Ans: } 0$$

$$3) \lim_{x \rightarrow 0} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right) \quad \text{Ans: } -\frac{1}{2}$$

$$4) \lim_{x \rightarrow 4} \left(\frac{1}{\ln(x-3)} - \frac{1}{x-4} \right) \quad \text{Ans: } \frac{1}{2}$$

part ii b

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$$

The limit of the given function is indeterminate form of the type $(\infty-\infty)$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\tan x} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{\cos x}{\sin x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\cos x - \cos 0}{x \sin x} \right) (\%)$$

By L'H

$$= \lim_{x \rightarrow 0} \left(\frac{\cos x + x \sin x - \cos 0}{\cos x + \sin x} \right) (\%)$$

by applying limit, we get

$$= \frac{\cos(0) + 0 \cdot \sin(0) - \cos(0)}{0 \cdot \cos(0) + \sin(0)}$$

By

L'H

$$= \lim_{x \rightarrow 0} \frac{-\sin x + \sin x + x \cos x + \cos x}{-x \sin x + \cos x + \cos x}$$

$$= \frac{0 + 0 + 0 + 0}{0 + 1 + 1}$$

$$= 0 \text{ Ans.}$$

3) The

 $\infty - \infty$

$$\lim_{x \rightarrow 0} \left(\frac{1}{e^x - 1} - \frac{1}{x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{x - e^x + 1}{x(e^x - 1)} \right)$$

By applying LHT;

$$= \lim_{x \rightarrow 0} \frac{1 - e^x + 0}{x(e^x - 0) + (e^x - 1)}$$

By applying LHT;

$$= \lim_{x \rightarrow 0} \frac{0 - e^x}{x e^x + e^x + 0}$$

By applying limit, we get-

$$= \frac{-e}{0 + 1 + 1} = \frac{-1}{2}$$

Date _____

Q.7 The ... $x \rightarrow 0$

$$\lim_{x \rightarrow 0} \left(\frac{1}{\ln(x-3)} - \frac{1}{(x-4)} \right)$$

$$= \lim_{x \rightarrow 4} \frac{x-4 - \ln(x-3)}{\ln(x-3)(x-4)}$$

$$\text{L.H.S.} = \lim_{x \rightarrow 4} \frac{1-0 - \frac{1}{(x-3)} \times 1}{\ln(x-3) + \ln(x-3) \times 1} \quad x \rightarrow 4$$

$$= \lim_{x \rightarrow 4} \frac{x-3-1}{(x-3) \ln(x-3) \times (x-3) + (x-4)}$$

$$= \lim_{x \rightarrow 4} \frac{x-3-1}{\ln(x-3) \times (x-3) + (x-4)}$$

By applying L.H.T,

$$= \lim_{x \rightarrow 4} \frac{1-0}{\ln(x-3) + (x-3) + 1} \quad \frac{0}{0+1}$$

$$= \lim_{x \rightarrow 4} \frac{1}{\ln(x-3) + 1 + 1}$$

$$= \lim_{x \rightarrow 4} \frac{1}{\ln(x-3) + 2}$$

$$= \frac{1}{2}$$

Maclaurin's Expansion

MACLAURIN'S

THEOREM

If $f(x)$ is n^{th} order at $x=0$; then the infinite sum of the polynomial

i.e. $f(x) = f(0) + f'(0)x + \dots$

EXAMPLE

PROBLEM

If $f(x)$ is n^{th} order at $x=a$, then it can be expanded as the infinite sum of the centered at $x=a$ i.e.:

$$f(x) = f(a) + f'(a)(x-a) + \dots$$

Maclaurin's & Taylor's Expansion:

MACLAURIN'S

THEOREM

If $f(x)$ is n^{th} order differentiable function at $x=0$; then it can be expanded as the infinite sum of the terms of the polynomial centred at $x=0$.
i.e.,

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \dots$$

TAYLOR'S

THEOREM

If $f(x)$ is n^{th} order differentiable function at $x=a$, then it can be expanded as the infinite sum of the terms of the polynomial centred at $x=a$.
i.e.:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \dots$$

Date _____

Q find the maclaurin's series of $f(x) = \sin x$

The maclaurin's series is given by

$$\begin{aligned}
 f(x) &= \sin x \\
 f(0) &= \sin(0) = 0 \\
 f'(0) &= \cos 0 \\
 f''(0) &= -\sin 0 \\
 f'''(0) &= -\cos 0 = -1 \\
 f^{(4)}(0) &= \sin 0 = 0 \\
 f^{(5)}(0) &= \cos 0 = 1 \\
 f^{(6)}(0) &= -\sin 0 = 0 \\
 f^{(7)}(0) &= -\cos 0 = -1 \\
 f^{(8)}(0) &= \sin 0 = 0
 \end{aligned}$$

Hence, the maclaurin's series is

$$\begin{aligned}
 \text{given by} \\
 \Rightarrow f(\sin x) &= 0 + x + 0 + \frac{(-1)x^3}{3!} + 0 + \frac{x^5}{5!} + 0 - \frac{x^7}{7!} + \dots
 \end{aligned}$$

$$\Rightarrow \sin x = \frac{x - x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\Rightarrow \sin x = \frac{x - x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$$

$$Q_2) f(x) = \ln(1-x)$$

$$\begin{aligned}
 f(x) &= \ln(1-x) \\
 f(0) &= \ln(1-0) \\
 f'(0) &= 0 \\
 \Rightarrow f'(x) &= \frac{-1}{1-x} \\
 f''(0) &= -1 \\
 \Rightarrow f''(x) &= \frac{-1}{(1-x)^2} \\
 f'''(0) &= -2 \\
 \Rightarrow f'''(x) &= \frac{-2}{(1-x)^3} \\
 f^{(4)}(0) &= -6 \\
 \Rightarrow f^{(4)}(x) &= \frac{-6}{(1-x)^4}
 \end{aligned}$$

$$\begin{aligned}
 \text{Q}_2) f(x) &= e^x \cos x \\
 f(x) &= \sin x \\
 f(x) &= \ln(1-x)
 \end{aligned}$$

Date _____

Q.2) $f(x) = \ln(1-x)$

$$f(x) = \ln(1-x)$$

$$f(0) = \ln(1-0)$$

$$f(0) = 0$$

$$\Rightarrow f'(x) = \frac{-1}{1-x}$$

$$f'(0) = -1$$

$$\Rightarrow f''(x) = \frac{(1-x)^2}{(1-x)^2}$$

$$f''(0) = -1$$

$$\Rightarrow f'''(x) = \frac{(1-x)^3}{(1-x)^3}$$

$$f'''(0) = -2$$

Q.2) $f(x) = e^x \cos x$

$$f(x) = \sinh x$$

$$f(x) = \ln(1+x)$$

$$f(x) = \sin^{-1}(x)$$

$$f^{(4)}(x) = \frac{-6}{(1-x)^4}$$

$$\Rightarrow f^{(4)}(0) = -6$$

Hence; the maclaurins expansion

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

Partial Differentiation:

Function of two or more variables:

There are some quantities which depend on two or more variables for example: The area of rectangle $\Rightarrow A = l \times b$ for a given pair of values of l and b , A has a definite value. The volume of parallelopiped $\Rightarrow V = l \times b \times h$ i.e. it depends on three variables: length, breadth and height.

A symbol Z which has a definite value for every pair of values of x and y is called a function of two independent variables x and y and we write $Z = f(x, y)$.

Partial Derivative

Let $Z = f(x, y)$ be a function of two variable x and y so if we keep y as constant and vary x alone then Z is a function of x only. The derivative of Z w.r.t x treating y as constant is called partial derivative of Z w.r.t x and is denoted by $\frac{\partial Z}{\partial x}$ or $\frac{\partial f}{\partial x}(x, y)$.